

AI-POWERED ERGONOMIC EXCELLENCE

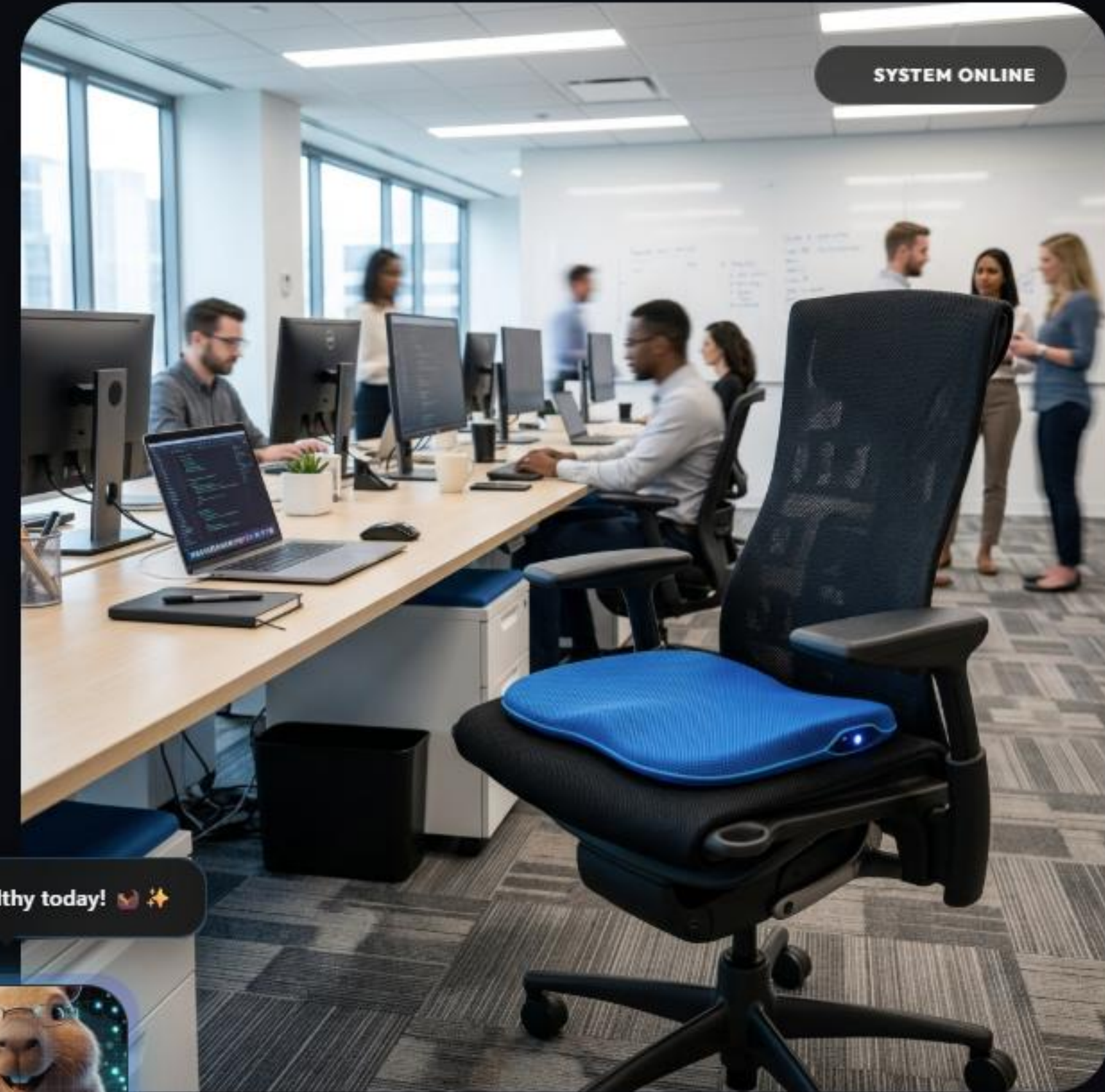
Improve your sitting with *Capybara Companion.*

Transform your sitting habits with our IoT smart cushion. Meet your AI health companion, detect posture in real-time, and embark on a gamified journey to spinal wellness.

What We Solve

Explore the Solution →

Hi! Let's sit healthy today! 🐾 ✨



Modern Office

Smart Classroom

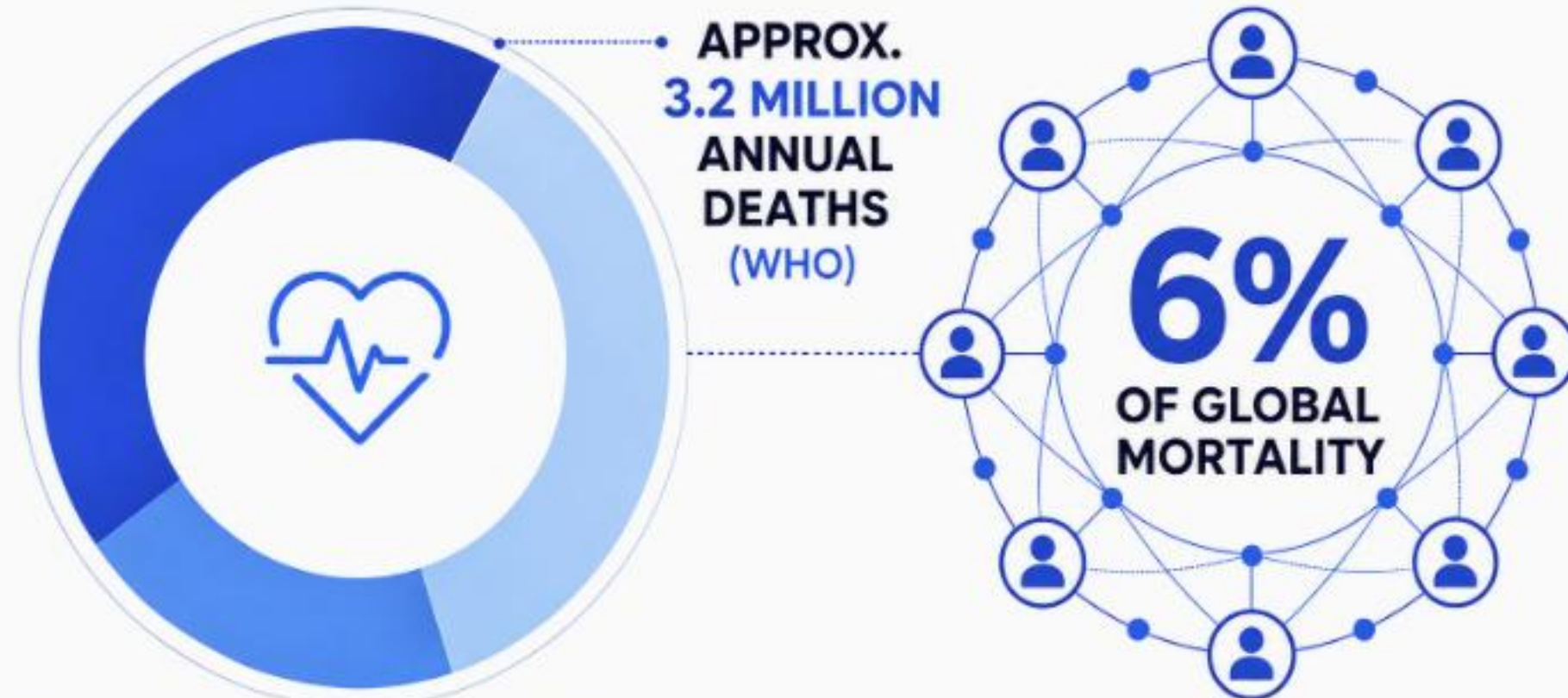
Home & Living

A Journey through *the Problem.*

Understanding the silent impact of sedentary lifestyles is the first step
toward a **revolutionary recovery.**



THE SILENT KILLER: GLOBAL MORTALITY



*Source: World Health Organization (WHO)



CHALLENGE 1

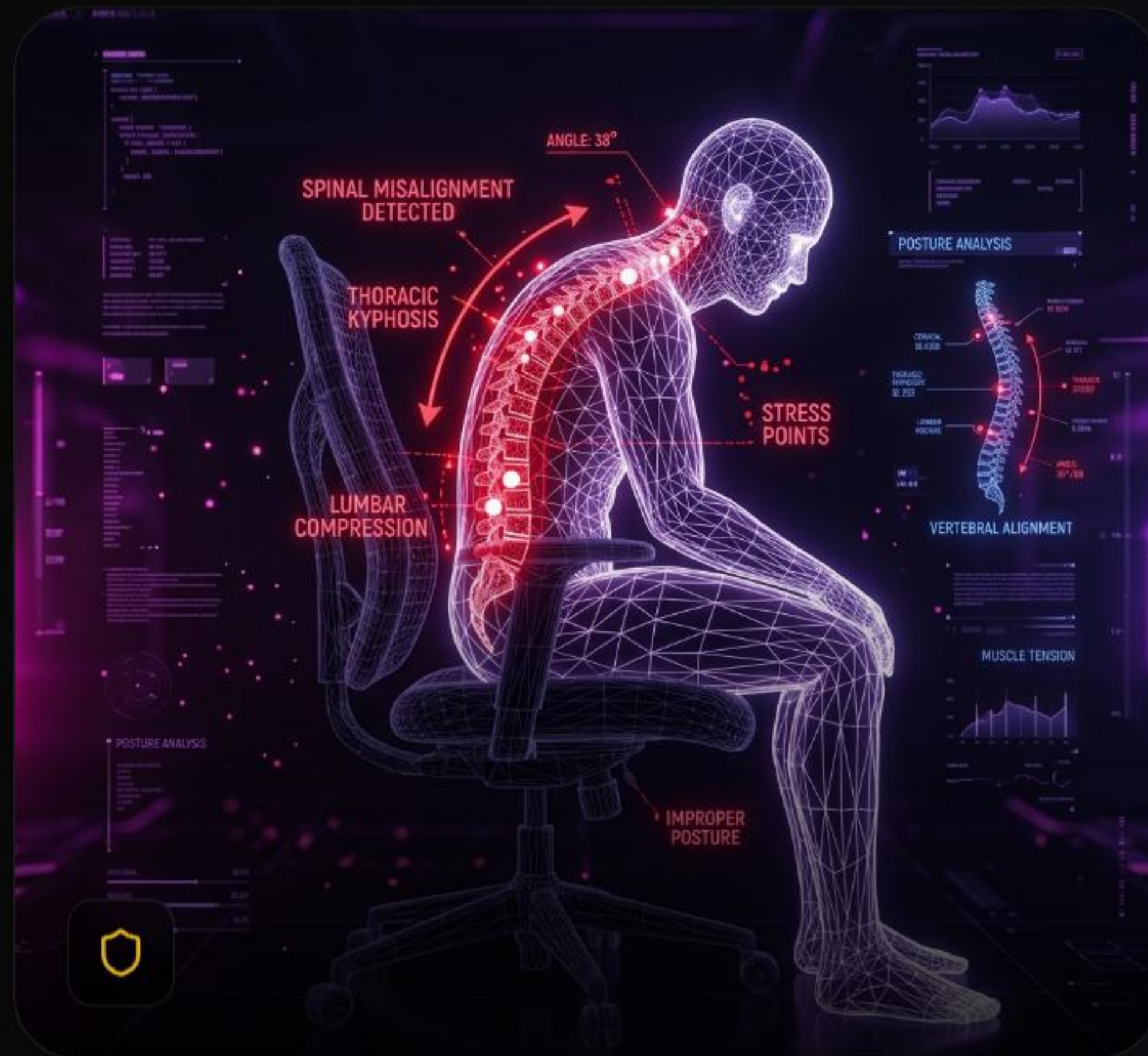
The Silent Killer

THE CRISIS

Physical inactivity and poor sitting postures account for approximately 3.2 million annual deaths globally, representing 6% of global mortality.

THE SOLUTION

Smart-Cushion proactively tracks your posture to break sedentary patterns and lower long-term health risks.



CHALLENGE 2

Spinal Misalignment

THE CRISIS

Improper sitting posture leads to thoracic kyphosis, lumbar compression, high muscle tension, and abnormal vertebral alignment (such as a 38-degree tilt angle).

THE SOLUTION

Smart-Cushion detects these bad sitting angles in real-time, alerting you to adjust and prevent spinal compression.



GLOBAL LOW BACK PAIN (LBP) BURDEN



Global Burden of Disease (GBD) Study



CHALLENGE 3

Rising Disease Trend

THE CRISIS

The global low back pain (LBP) burden is rising rapidly, increasing from 619 million cases in 2020 to a projected 843 million cases by 2050.

THE SOLUTION

Smart-Cushion helps you maintain spinal health continuously, preventing you from becoming part of this growing global statistic.



Check our *solution.*

Join thousands of office professionals who have
already reclaimed their spinal health.

[View Our Product](#)

Our technology is *more than just hardware.*
**It's a comprehensive ecosystem
designed to improve your health.**



3D PARALLAX



HOVER TO TILT



THE MASTERPIECE

Design *DNA*

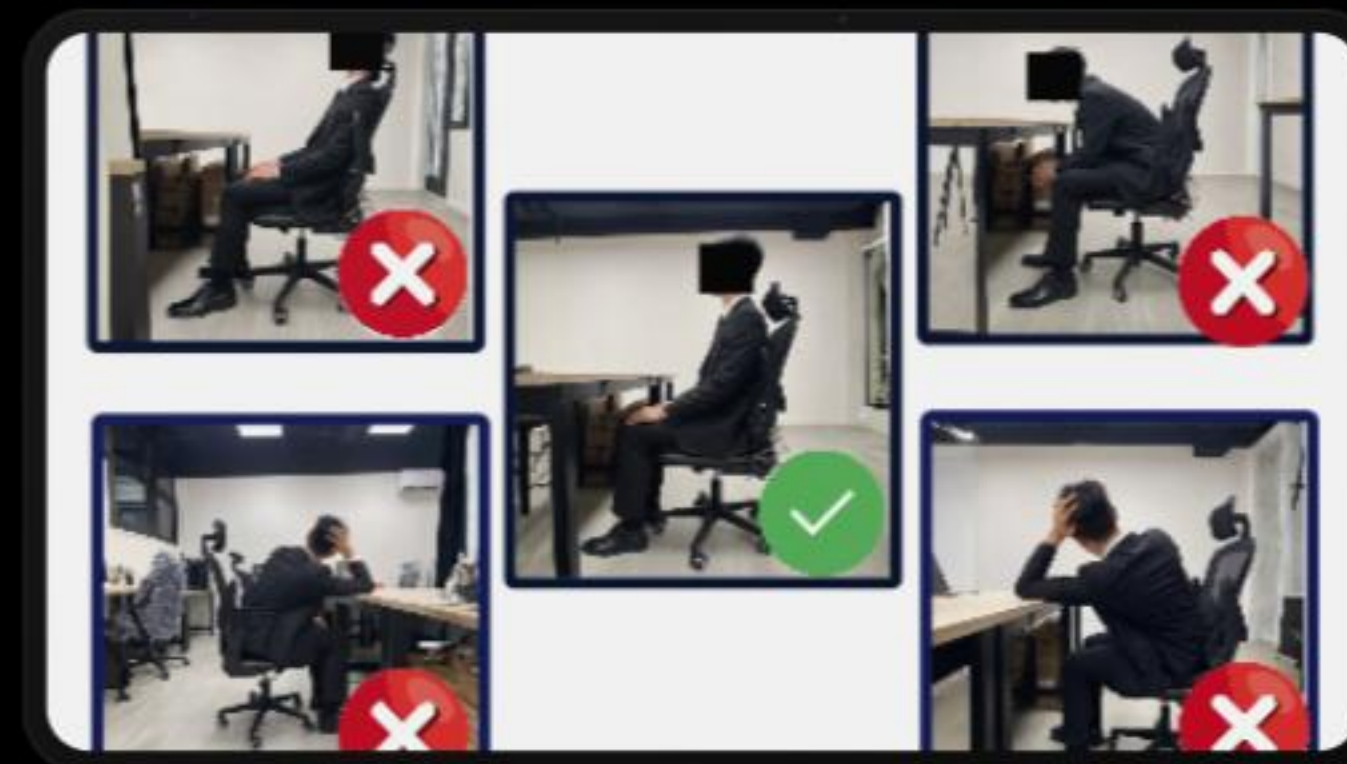
A perfect blend of aesthetic elegance and technological power.



TECH / 01

5-Posture Recognition

Our advanced AI analyzes data from a high-precision 3x3 sensor matrix to accurately identify 5 distinct sitting postures.

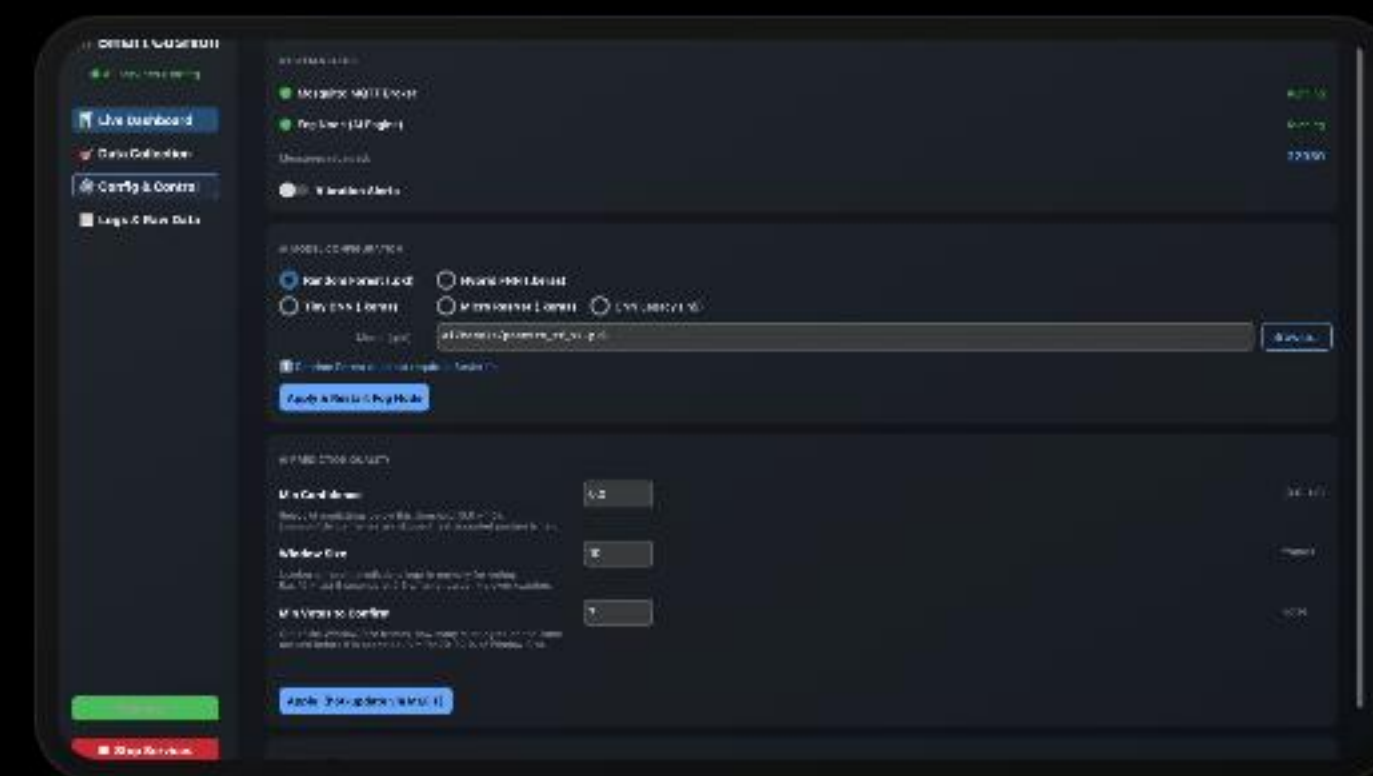




TECH / 02

Embedded Fog Computing

Localized data processing ensures zero latency and maximum privacy for your sensitive health data.





TECH / 03

Real-time App Feedback

Receive instant posture corrections and long-term health trends directly through our dedicated mobile application.

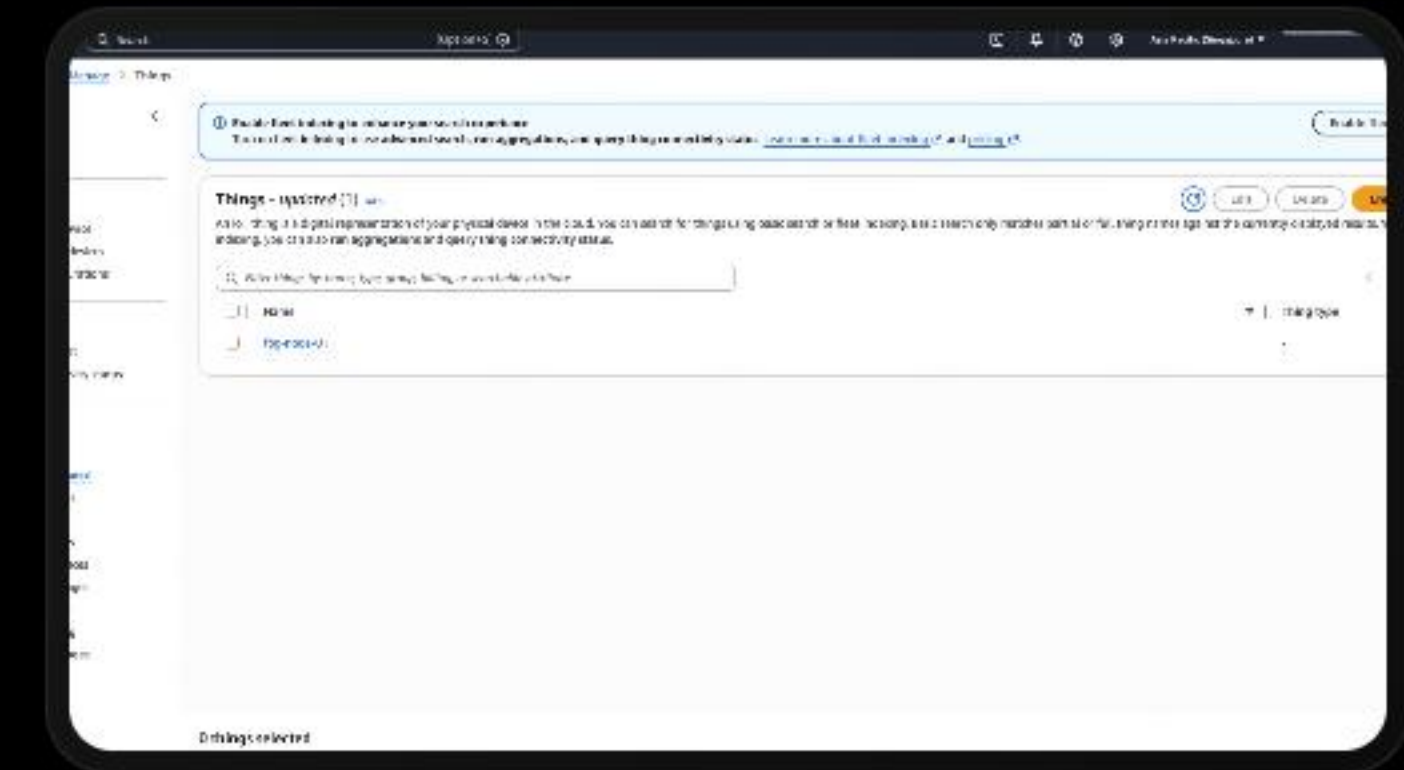




TECH / 04

Cloud Integration

Built on a serverless AWS stack using IoT Core, Lambda (Python), API Gateway, and Amazon DynamoDB for real-time WebSocket sync and historical session storage.

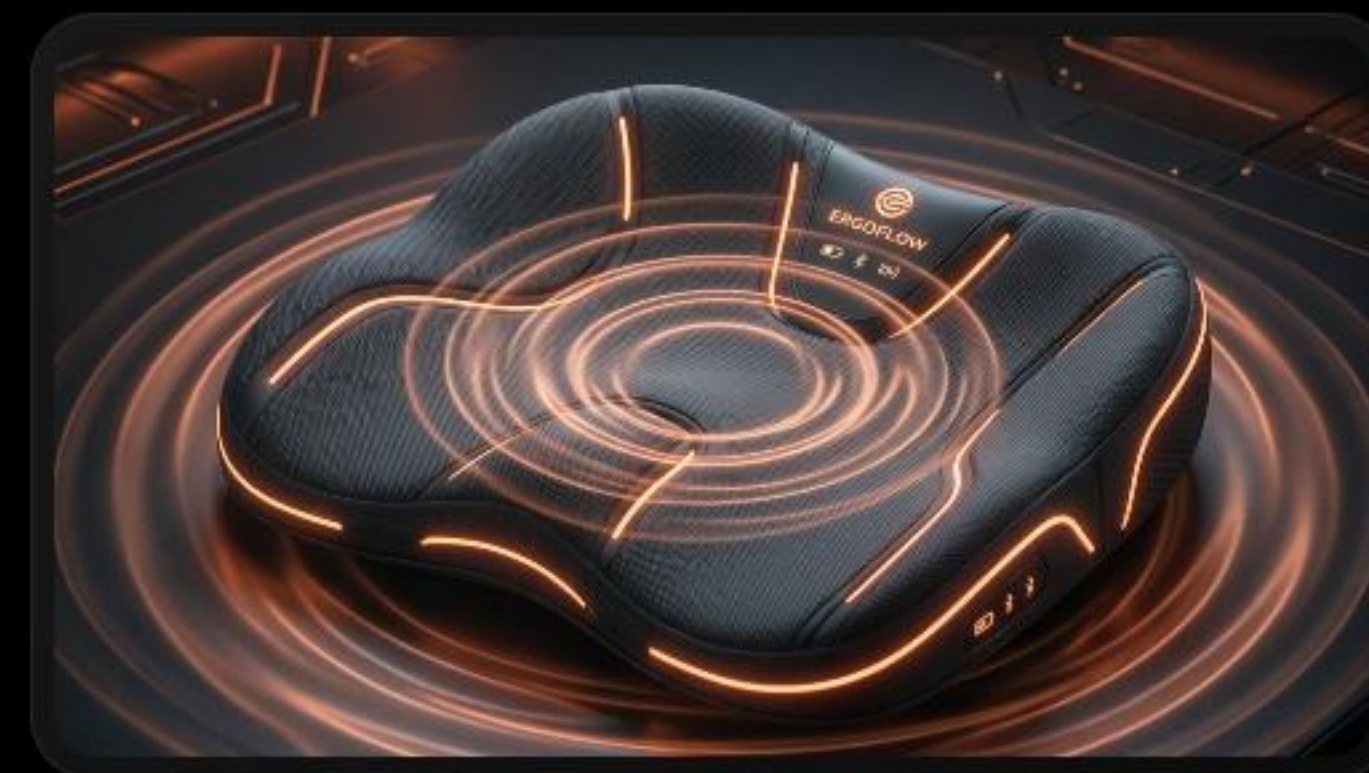




TECH / 05

Smart Vibrations

Subtle haptic feedback alerts you immediately when your posture needs adjustment without distraction.





TECH / 06

Capybara Data Journey

Observe your real-time posture through your virtual Capybara companion. A healthy spine keeps your buddy happy and thriving!





See how it *all connects.*

Explore Architecture →

 Technical Documentation

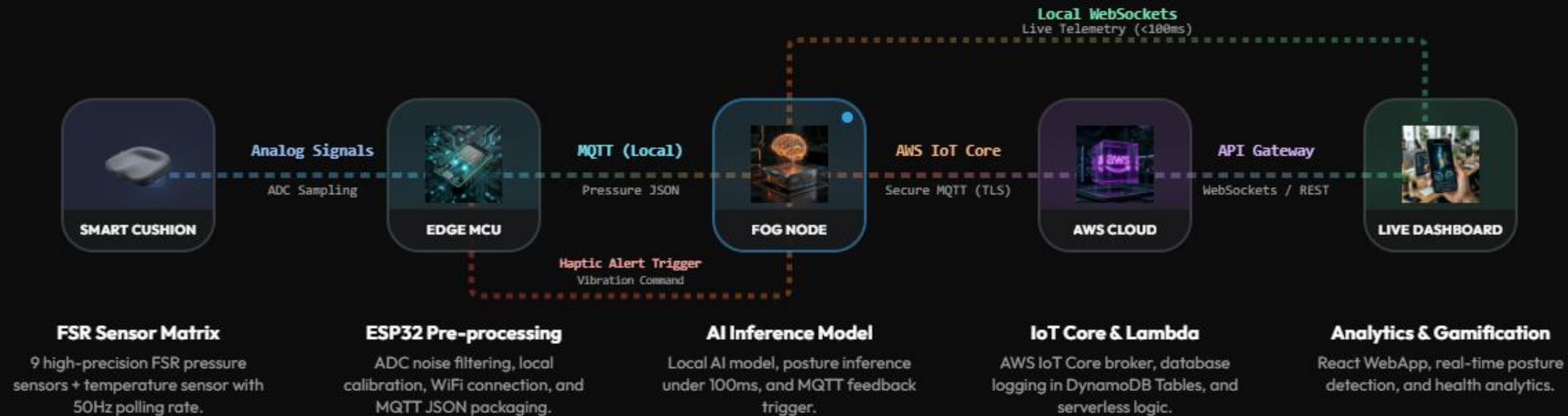
System *Architecture* Models

A comparative view of standard architectural models applied to the Smart Cushion ecosystem.



● PRACTICAL IMPLEMENTATION

From *Sensor* to *Screen*



CLOUD INTEGRATION ECOSYSTEM

AWS Serverless & *IoT Infrastructure*

How we process posture sensor logs, manage real-time WebSocket sessions, and synchronize sitting statistics with low latency.



100% Free-Tier Compatible

AWS serverless triggers cost \$0/month under academic tier quotas.

[Live Posture Dashboard](#) →



AWS IoT Core

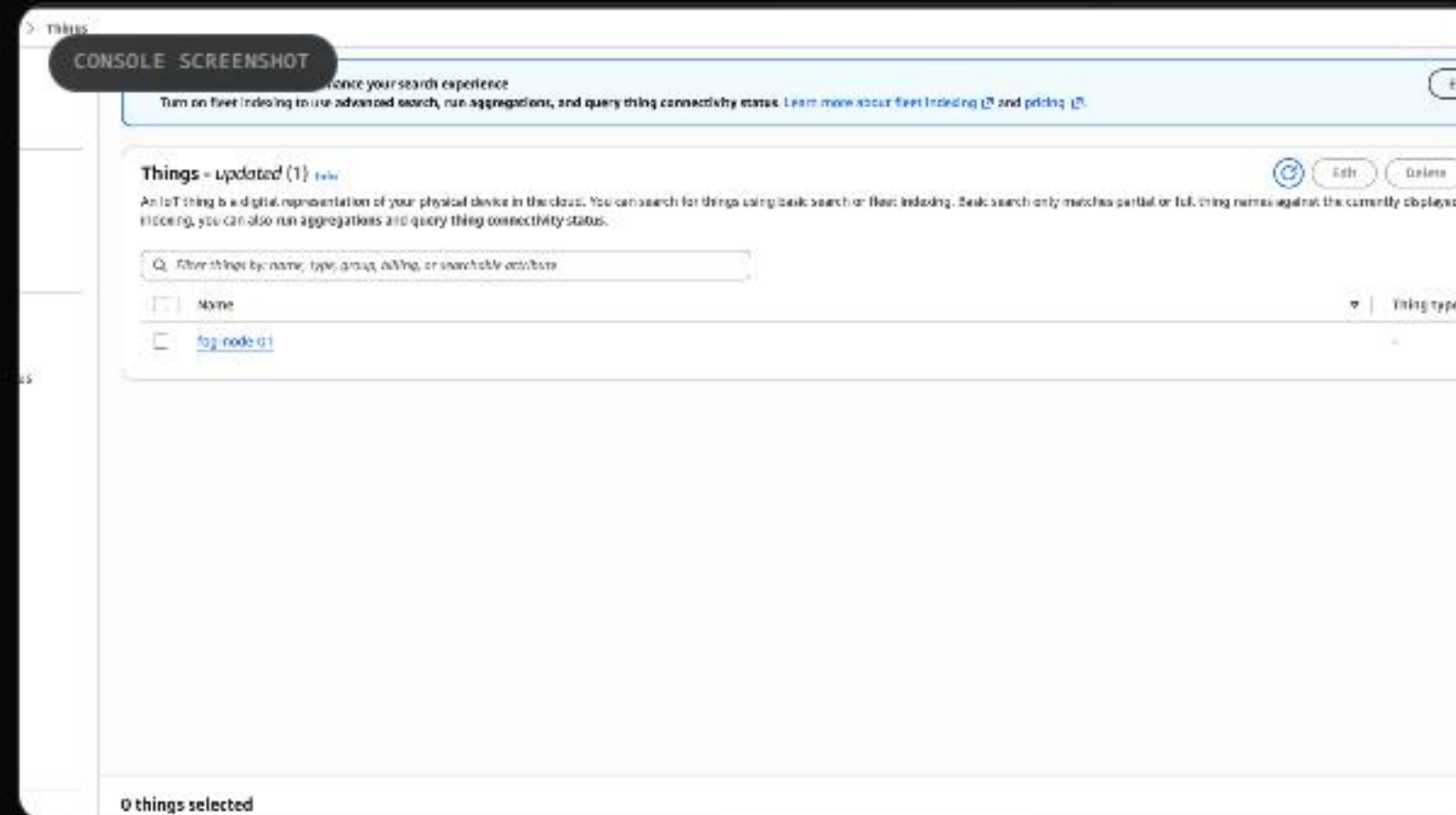
SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Connects local Fog Nodes (e.g. 'fog-node-01') securely to the cloud via TLS-encrypted MQTT. It filters messages via IoT Rules to trigger specialized Lambda workflows asynchronously.

ACTIVE DEPLOYMENTS

- Thing: fog-node-01
- Topic: cushion/+/summary (Session statistics)
- Topic: cushion/+/event (Alerts and anomalies)



> IoT Rule SQL Query

ACTIVE

```
SELECT * FROM 'cushion/+/summary'

-- Action: Invoke Lambda Function
-- Target: SmartCushion-ProcessSummary
```



100% Free-Tier Compatible

AWS serverless triggers cost \$0/month under academic tier quotas.

Live Posture Dashboard →



AWS CloudFormation

SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Deploys and governs our entire smart cushion architecture. By using SAM templates, CloudFormation spins up the DynamoDB tables, REST APIs, WebSocket routes, and Lambda functions with strict least-privilege IAM roles automatically.

ACTIVE DEPLOYMENTS

- Stack: smart-cushion-cloud
- IAM Roles: LambdaExecutionRole, ApiGatewayPushDep
- Outputs: WebSocketURI, RestApiGatewayURL

CONSOLE SCREENSHOT

Filter status: Active View nested

Stack name	Status	Created time	Description
smart-cushion-cloud	UPDATE_COMPLETE	2020-04-20 15:12:16 UTC+0200	Smart Cushion Cloud Backend + Lambda + API Gateway (REST + WebSocket)
aws-sam-cli-managed-default	CREATE_COMPLETE	2020-04-20 15:14:50 UTC+0200	Managed Stack for AWS SAM CLI

>_ CloudFormation Event Stack

ACTIVE

```
CREATE_COMPLETE AWS::DynamoDB::Table UsersTable
CREATE_COMPLETE AWS::Serverless::Function GachaRoll
CREATE_COMPLETE AWS::ApiGateway::RestApi ServerlessRestApi
```



100% Free-Tier Compatible

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Live Posture Dashboard →



Amazon CloudWatch

SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Collects system performance and runtime metrics.
Monitors lambda cold-starts, captures application execution exceptions, tracks active user WebSocket connections, and triggers administrative alerts if anomaly detection limits are crossed.

ACTIVE DEPLOYMENTS

- Log Group: /aws/lambda/SmartCushion-AddGems
- Metric Filter: GoodPostureRatio
- Alarm: HighLatencyThresholdAlert

CONSOLE SCREENSHOT - new Summary

Get-set-go with unified data store

Log groups (8)

By default, we only load up to 10,000 log groups.

Filter log groups or by pattern search

Filter metrics

Log group	Log class	Anomaly d...	Detection pr...	Devenir tak...	Data pro...	Sensitivity...	Retention	Account n...
/aws/lambda/SmartCushion-AddGems	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-GetGall...	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-GetGallySummary	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-GetSeedans	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-GetSeednfo	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-Login	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-ProcessEvent	Standard	Configure	Off	Off	-	-	Never expire	-
/aws/lambda/SmartCushion-ProcessSummary	Standard	Configure	Off	Off	-	-	Never expire	-

> CloudWatch Alarm Log

ACTIVE

[INFO] 2026-05-21 17:07:13 - Posture summary processed for peter_ntust
[WARN] 2026-05-21 17:08:00 - Lambda execution duration exceeded 2000ms
[INFO] 2026-05-21 17:08:15 - Active connection count: 24



100% Free-Tier Compatible

AWS serverless triggers cost \$0/month under academic tier quotas.

Live Posture Dashboard →



Amazon DynamoDB

SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Stores all project entities. Uses high-performance Partition Keys to handle parallel stream logging, keeping user profiles, sessions, and WebSocket links synchronized in real-time.

ACTIVE DEPLOYMENTS

- smart-cushion-users (Account progress & gems)
- smart-cushion-sessions (Detailed posture logs)
- smart-cushion-daily-summary (Pre-calculated stats)
- smart-cushion-ws-connections (WebSocket mapping)

CONSOLE SCREENSHOT

home	Schema	Partition key	Sort key	Indexes	Replication Regions	Deletion protection	Favorite	Read capacity mode
smart-cushion-daily-summary	Active	date_id (S)	date (S)	0	0	Off	☆	On-demand
smart-cushion-sessions	Active	device_id (S)	timestamp (S)	0	0	Off	☆	On-demand
smart-cushion-users	Active	username (S)	-	0	0	Off	☆	On-demand
smart-cushion-ws-connections	Active	connection_id (S)	-	0	0	Off	☆	On-demand

>_ DynamoDB Item: smart-cushion-users

ACTIVE

```
{
  "username": "peter_ntust",
  "gems": 250,
  "unlocked_items": ["copybara_gold", "desk_avatar_02"],
  "streak_days": 5,
  "last_active": "2026-05-21"
}
```



100% Free-Tier Compatible

AWS serverless triggers cost \$0/month under academic tier quotas.

Live Posture Dashboard →



AWS Lambda

SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Executes business logic on-demand without server overhead. Manages user authentication, daily posture summaries, and runs our anti-cheat reward logic.

ACTIVE DEPLOYMENTS

- SmartCushion-ProcessSummary (Stores MQTT reports)
- SmartCushion-AddGems (Postures-to-gems engine)
- SmartCushion-GachaRoll (Anti-cheat gacha logic)
- SmartCushion-GetDailySummary (Speedy API backing)

CONSOLE SCREENSHOT

Serverless generative AI webinar series

Serverless generative AI webinar series for a hands-on workshop on Agents AI, Intelligent Document Processing, and building real-time APIs for drubots.

Functions (11)

Search by attributes or search by keyword

☐	Function name	Description	Package type	Runtime	Type	Last modified
☐	SmartCushion-GetUserInfo	-	Zip	Python 3.11	Standard	2 weeks ago
☐	SmartCushion-GetDailySummary	-	Zip	Python 3.11	Standard	3 weeks ago
☐	SmartCushion-AddGems	-	Zip	Python 3.11	Standard	2 weeks ago
☐	SmartCushion-ProcessSummary	-	Zip	Python 3.11	Standard	2 days ago
☐	SmartCushion-Login	-	Zip	Python 3.11	Standard	2 weeks ago
☐	SmartCushion-ProcessEvent	-	Zip	Python 3.11	Standard	3 weeks ago
☐	SmartCushion-GetGems	-	Zip	Python 3.11	Standard	3 weeks ago
☐	SmartCushion-GetDailySummary	-	Zip	Python 3.11	Standard	2 days ago
☐	SmartCushion-GetMessage	-	Zip	Python 3.11	Standard	3 weeks ago
☐	SmartCushion-GetUserRole	-	Zip	Python 3.11	Standard	2 weeks ago
☐	SmartCushion-GetConnect	-	Zip	Python 3.11	Standard	3 weeks ago

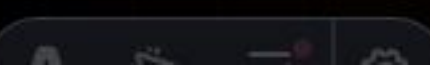
> SmartCushion-AddGems Lambda

ACTIVE

```
def lambda_handler(event, context):
    user = db.get_user(event['username'])
    nup_duration = event['nup_duration']

    # Securely calculate gems: 1 min NUP = 5 Gems
    earned_gems = int(nup_duration / 60) * 5
    db.update_gems(user, earned_gems)

    return {"status": "success", "earned": earned_gems}
```





AWS API Gateway

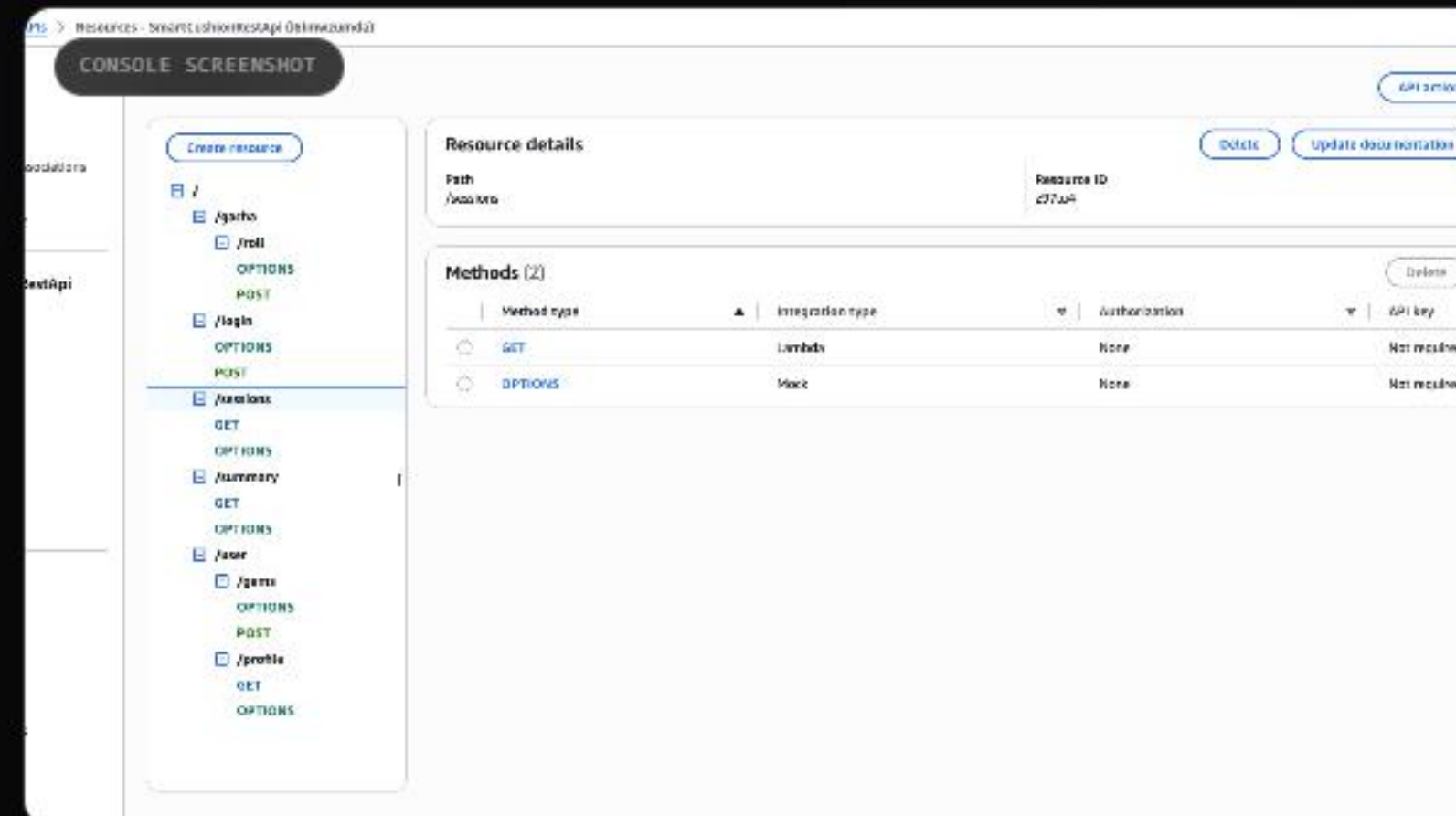
SERVERLESS INFRASTRUCTURE

CORE PURPOSE & RESPONSIBILITY

Acts as the portal to the cloud. Integrates directly with the React webapp via REST API and acts as a stateful WebSocket proxy to relay live posture alerts from the physical cushion.

ACTIVE DEPLOYMENTS

- REST API: /login, /sessions, /summary
- Progression & Reward Endpoints: /user/gems, /gacha/roll
- WebSocket Route: \$connect, \$disconnect, \$default



> API Route Integration Map

ACTIVE

```
POST /gacha/roll -> Lambda: SmartCushion-GachaRoll
GET /summary -> Lambda: SmartCushion-GetDailySummary
WSS /live -> WebSocket Proxy API
```



100% Free-Tier Compatible

AWS serverless triggers cost \$0/month under academic tier quotas.

Live Posture Dashboard →

CONTROL CENTER

Your Health, Improved alongside *Capybara.*

The PostureAI Dashboard provides a comprehensive suite of tools to monitor and analyze your sitting habits in real-time.

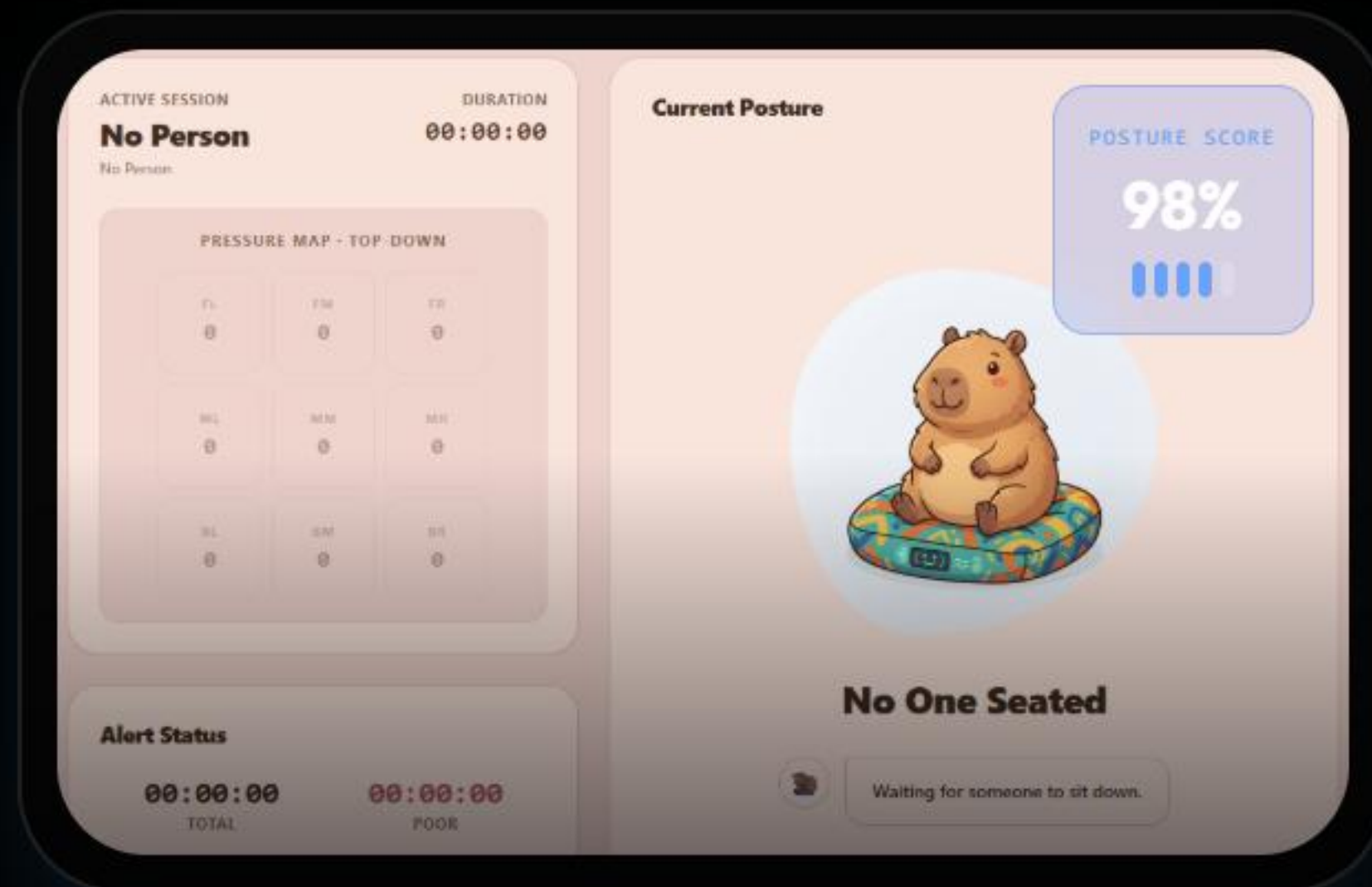




01 / DASHBOARD VIEW

Real-time Monitoring

Experience zero-latency posture tracking with our 9-point pressure heatmap. Monitor sensor values and system status instantly.

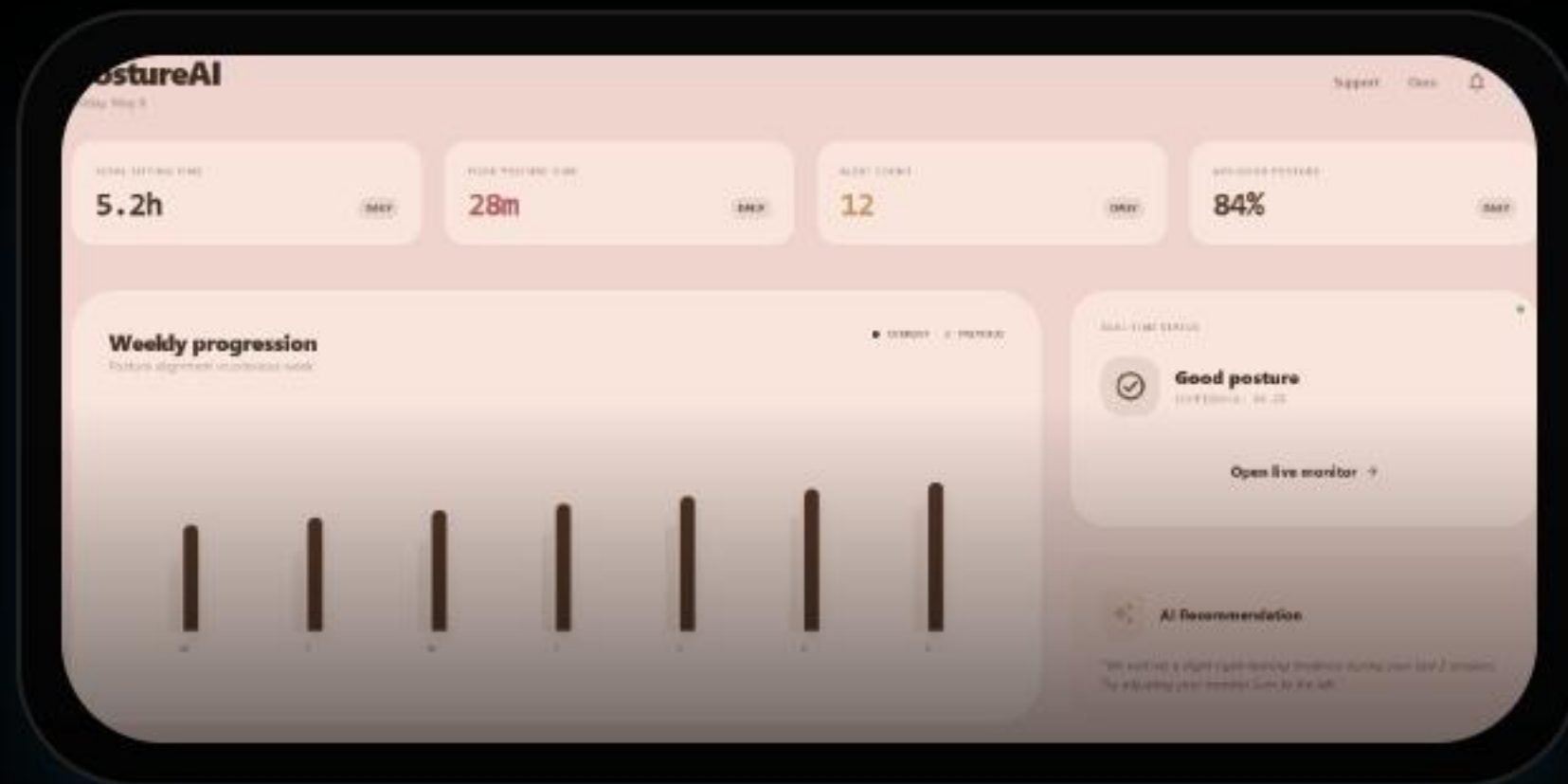




02 / DASHBOARD VIEW

Weekly Progression

Track your long-term alignment trends and see how your posture improves over time with detailed weekly analytics.





03 / DASHBOARD VIEW

Performance Analysis

Deep dive into your sitting habits. Identify your best and worst days to make data-driven health decisions.





Session History

Last 30 days · device cushion-01

Export

PDF

DATE	DURATION	GOOD %	POOR (TIMES)	ALERTS	
May 06, 10:03	00:00:34	82%	01:02:05	1 Warn	3
May 06, 09:55	00:02:23	100%	01:02:01	0 Warn	0
May 06, 09:54	00:00:50	34%	00:02:35	0 Warn	0
May 06, 09:39	00:02:00	74%	00:02:14	0 Warn	0

04 / DASHBOARD VIEW

Session History

Review every sitting session in detail. Export logs for medical consultations or personal health tracking.





05 / DASHBOARD VIEW

Capybara Gacha

Earn points by maintaining good posture and use them in our Gacha system to unlock rare Capybara skins and accessories.





Capy Collection

BROWSE YOUR STICKER LIBRARY

COMPLETION

4 / 9



Slouch



Bandage



Tired



Sparkles



???

KEEP SPINNING!



???

KEEP SPINNING!



???

KEEP SPINNING!



???

KEEP SPINNING!

06 / DASHBOARD VIEW

Avatar Collection

Browse and equip your unlocked Capybara themes. Show off your dedication to spinal health with exclusive digital pets.

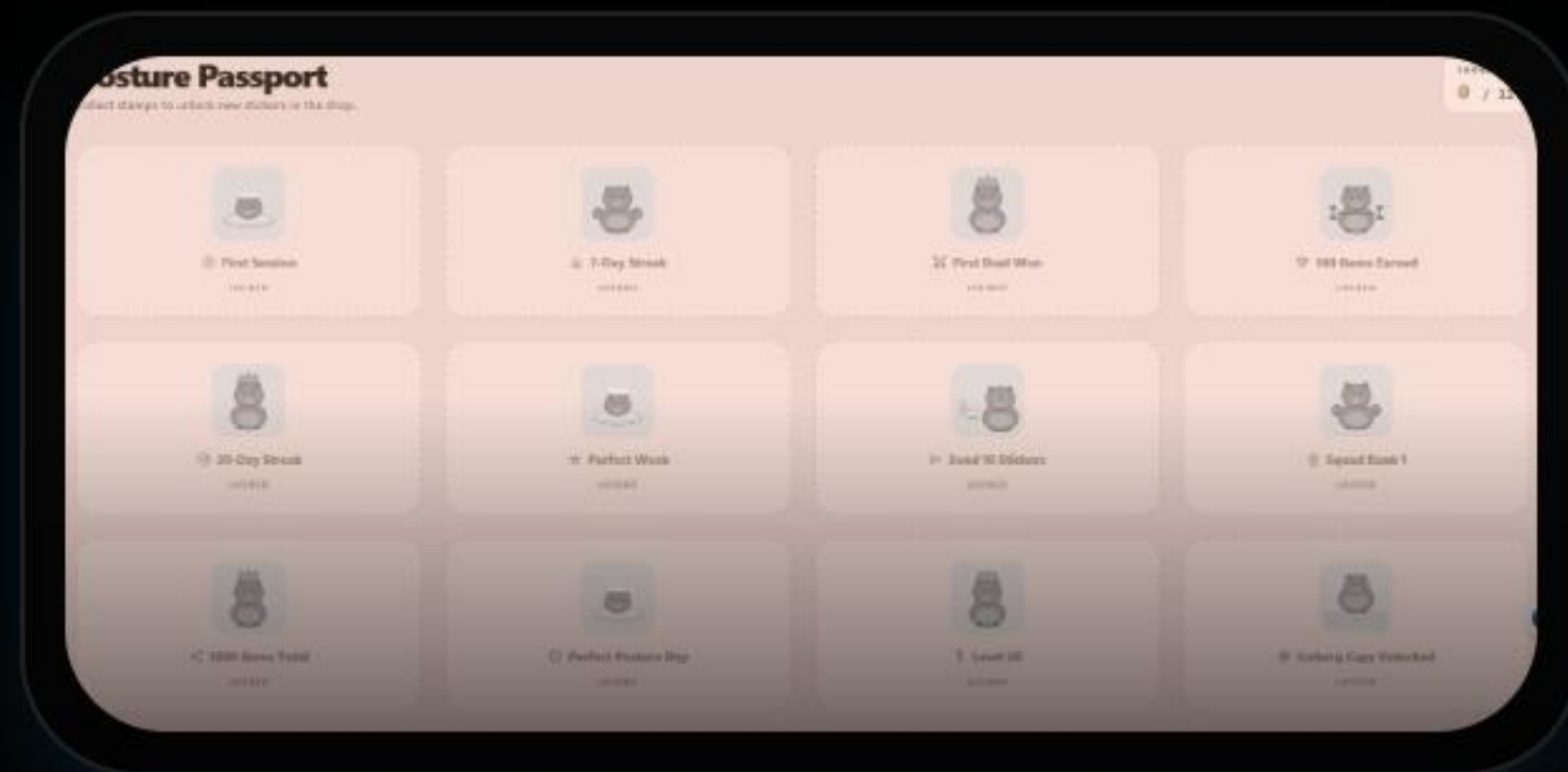




07 / DASHBOARD VIEW

Health Passport

Your unified ergonomic profile. Track achievements, daily streaks, and overall wellness milestones in your Capybara Passport.





SmartCushion

[Introduction](#)

[What we solve](#)

[Product](#)

[Architecture](#)

[Cloud](#)

[Dashboard](#)

[Team](#)

[Contact Us](#)

[Future Work](#)

Take Command of *Your Wellbeing.*

[Try Demo](#)

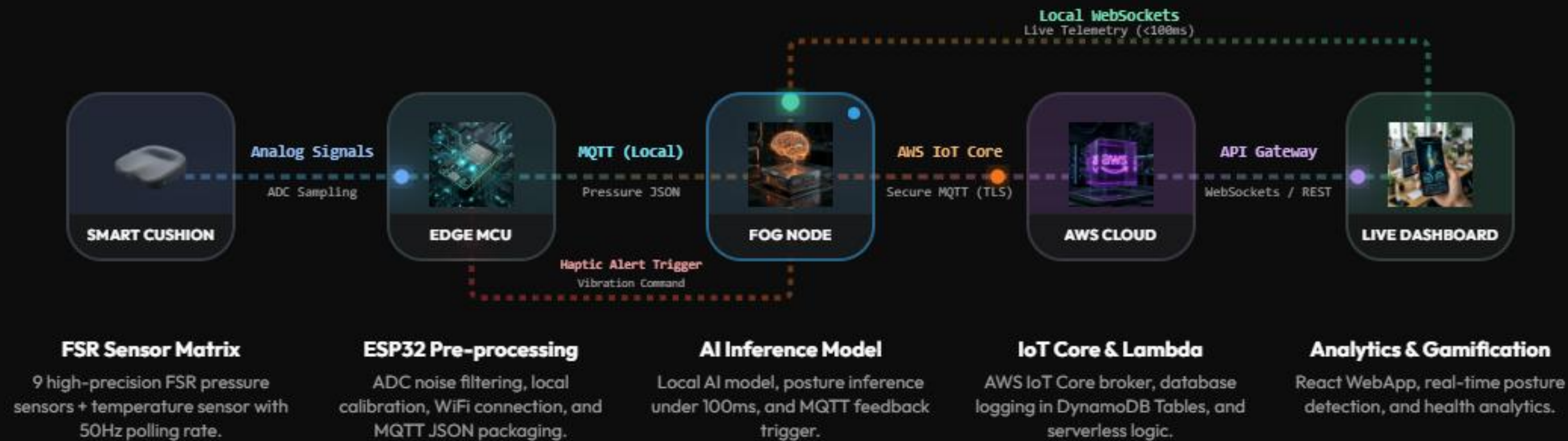
EXPERIENCE THE LIVE DATA FLOW

● SCALABLE ARCHITECTURE

From *Sensor* to *Screen*

🔗 SCALE SYSTEM ARCHITECTURE ➔

Explore Cloud Engine

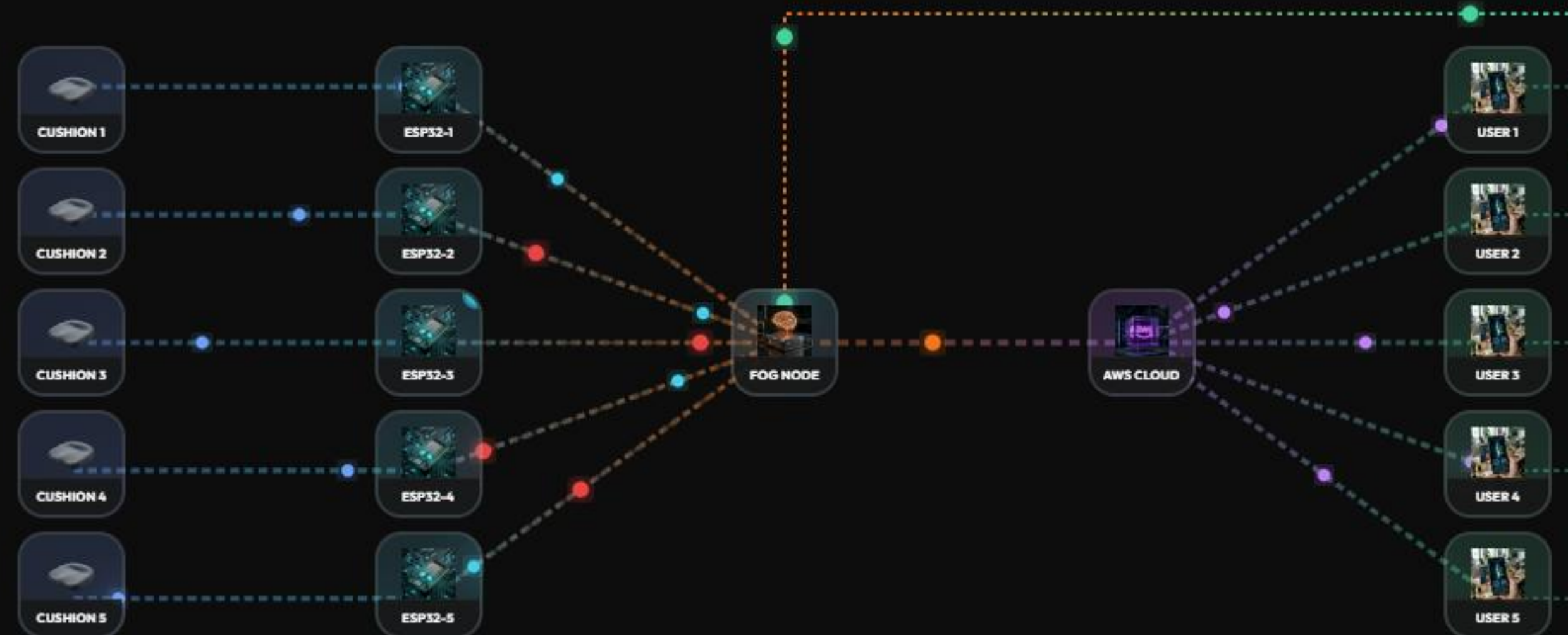


[SCALABLE ARCHITECTURE](#)

Scale *Edge Clusters* to *Dashboards*

[↩ RESET TO STANDARD VIEW](#)

Click the highlighted layers to view scaling strategies.



The Minds Behind *Smart Cushion*

Meet the innovators dedicated to revolutionizing ergonomic health through
AI and IoT technology.



**To Nguyen Tan
Phuong**

HARDWARE & EDGE
ENGINEER

Hardware architecture specialist. Phuong focuses on sensor integration and optimizing edge computing...



Tran Viet Nam

FOG & HARDWARE
INTEGRATION

Architects scalable local infrastructures and bridges the gap between hardware sensors and cloud services.



Nguyen Thao Huong

AI CONSULTANT & MODEL
DESIGNER

AI expert specializing in specialized model design and optimizing neural network architectures for health...



Dong Boi Thi

CLOUD & DASHBOARD
DEVELOPER

Specializes in cloud backend frameworks and real-time data visualization systems, ensuring seamless data flow.



Hoang Mai Vu

REAL-TIME DASHBOARD
ARCHITECT

Architects high-performance real-time dashboards, specializing in seamless data visualization and sub-100ms...

Simple, Transparent Pricing

Choose the plan that fits your lifestyle. All cushions come with a 2-year warranty and lifetime app access.

Smart Cushion Solo

Perfect for home office setups.

\$199 / one-time

- ✓ 1x Smart Cushion
- ✓ Mobile App Access
- ✓ Cloud Backup
- ✓ 2-Year Warranty

Buy Now

POPULAR

Smart Cushion Bundle

For home and office use.

\$349 / one-time

- ✓ 2x Smart Cushion
- ✓ Family App Sharing
- ✓ Priority Support
- ✓ Lifetime Warranty

Buy Now

NTUST FINAL PROJECT

Get in Touch with *Our Team.*

Have questions about our Cloud-Fog architecture or AI implementation? Reach out to the researchers behind Smart Cushion.



Location

National Taiwan University of Science and Technology (NTUST)
No. 43, Keelung Rd, Sec. 4, Da'an Dist, Taipei City, Taiwan.



Course Info

Course: Cloud Fog Computing (114-2)
Category: Smart IoT Spinal Wellness Solution



GitHub Hub

Our web repository acts as the primary hub. The README provides structured system architecture mappings and direct paths to the ESP32 Firmware, MQTT Fog Node, and serverless AWS Cloud stacks.

[Explore Ecosystem →](#)



Project Inquiry

GENERAL EMAIL

 T14cloudfog_team5_smartcushion@gmail.com

LAB LOCATION

 C217, Hwa Hsia Campus, TaiwanTech

Technical Discussion?

We are highly open to academic feedback regarding our real-time posture classification latency and AWS IoT core MQTT benchmark metrics.

[Send us a Message](#)